

Problem 15.21

If $d^{(12)} = 12\%$, calculate the accumulated value of 100 paid at the end of each month for 12 months.

Solution

The nominal annual discount rate convertible monthly is

$$d^{(12)} = 0.12.$$

Thus, the monthly discount rate is

$$d = \frac{0.12}{12} = 0.01.$$

The relationship between the effective monthly interest rate i and the monthly discount rate d is

$$1 + i = \frac{1}{1 - d}.$$

Therefore,

$$i = \frac{1}{1 - 0.01} - 1 = \frac{1}{99}.$$

The accumulated value of 100 paid at the end of each month for 12 months is

$$100s_{\overline{12}|i}.$$

Using

$$s_{\overline{n}|i} = \frac{(1 + i)^n - 1}{i},$$

we get

$$\begin{aligned} AV &= 100 \left(\frac{\left(1 + \frac{1}{99}\right)^{12} - 1}{\frac{1}{99}} \right) \\ &\approx 1268.9632. \end{aligned}$$

Hence, the accumulated value is

$$\boxed{1268.96}.$$